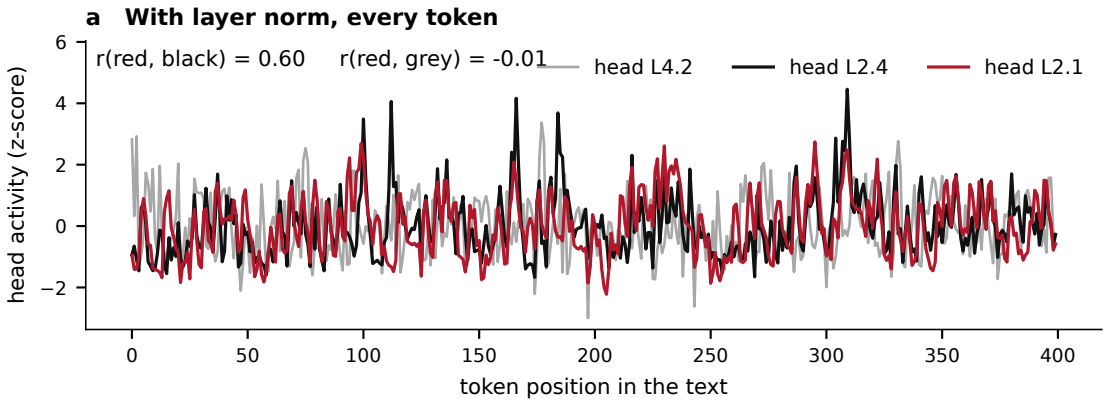
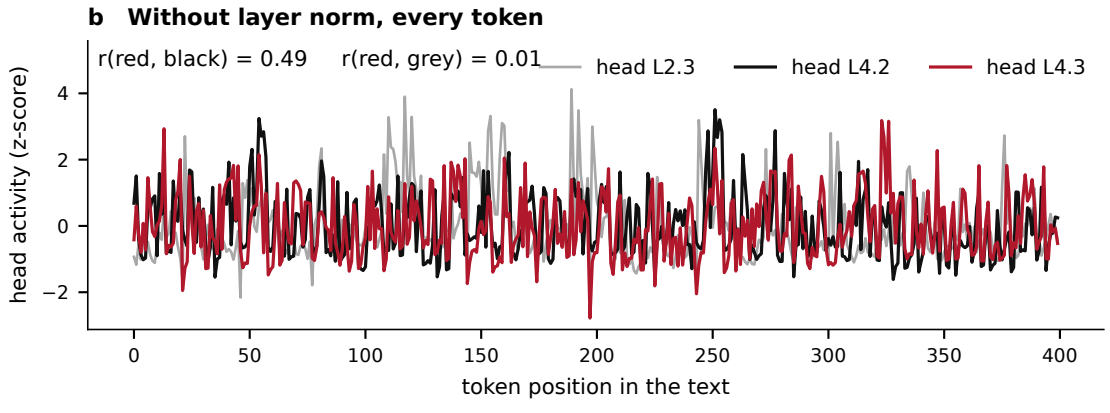
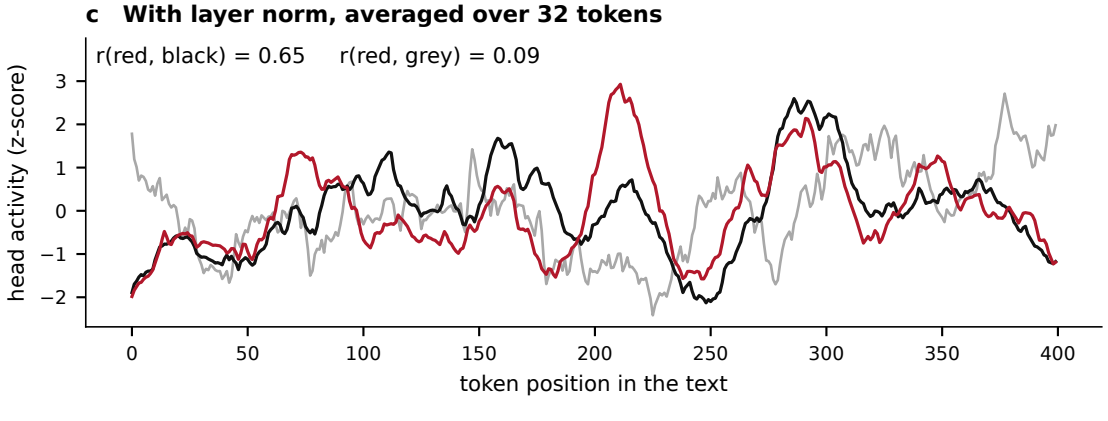
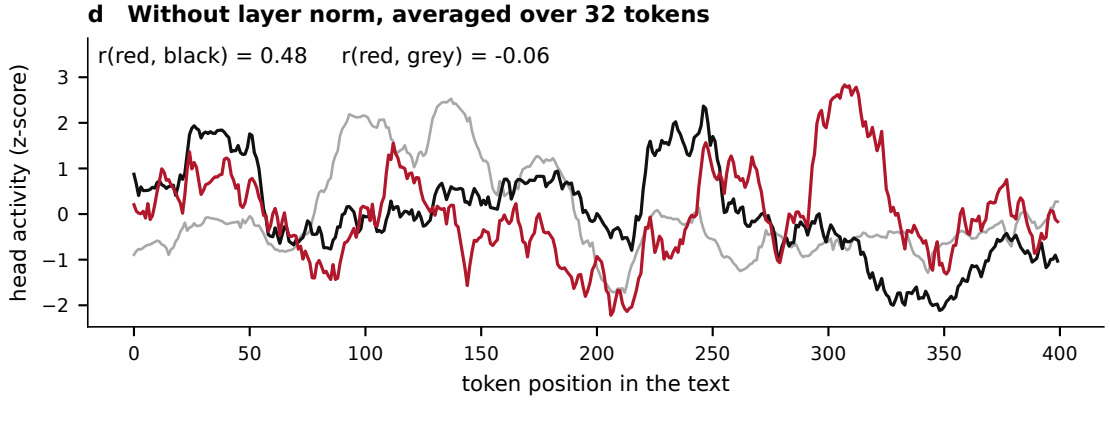
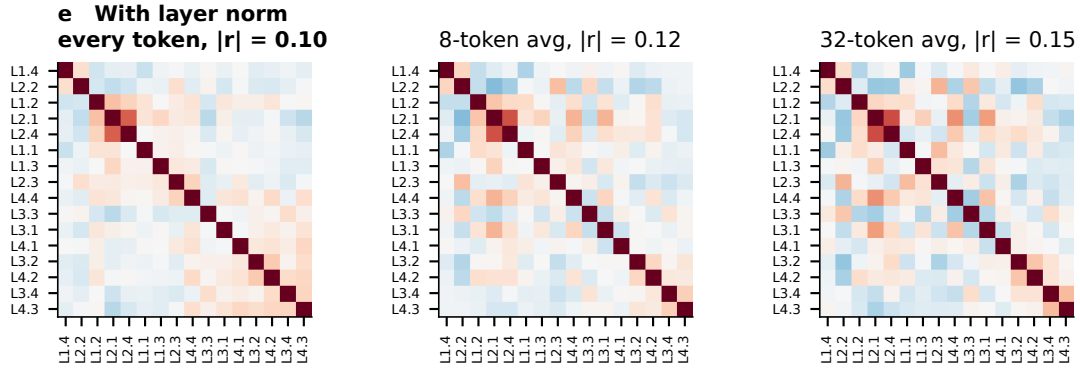
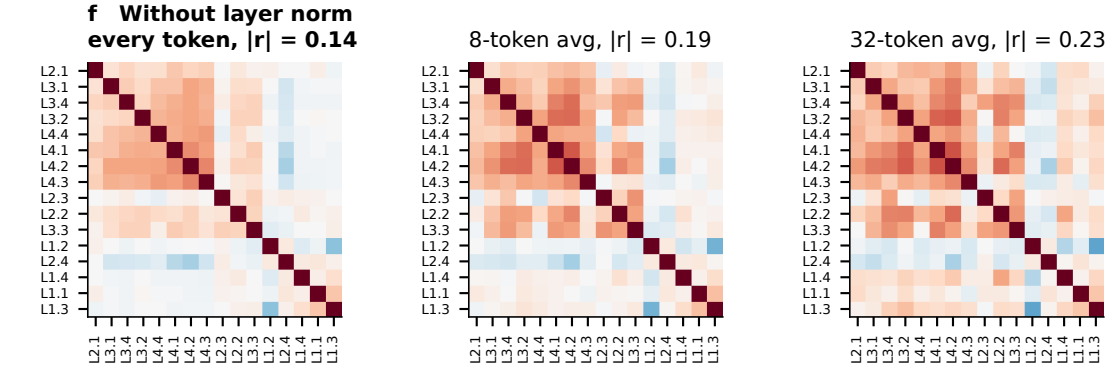

a With layer norm, every token
r(red, black) = 0.60
r(red, grey) = -0.01
head L4.2
head L2.4
head L2.1
head activity (z-score)
token position in the text

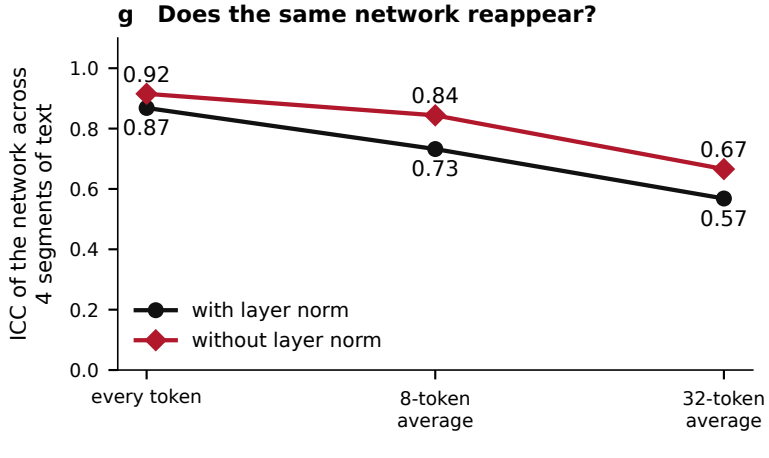

b Without layer norm, every token
r(red, black) = 0.49
r(red, grey) = 0.01
head L2.3
head L4.2
head L4.3
head activity (z-score)
token position in the text

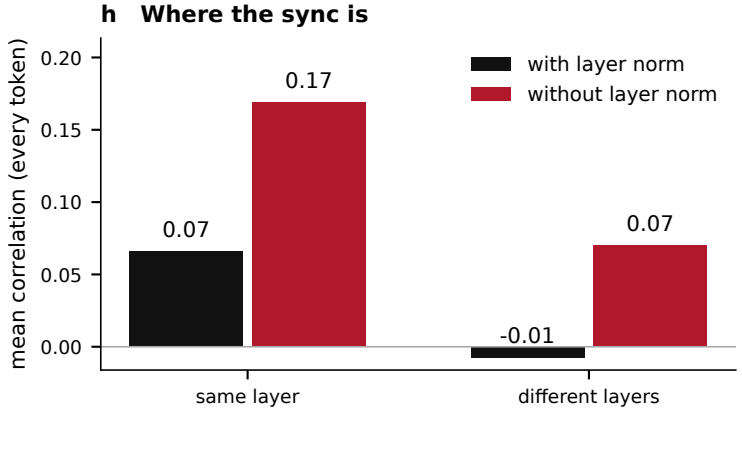

c With layer norm, averaged over 32 tokens
r(red, black) = 0.65
r(red, grey) = 0.09
head activity (z-score)
token position in the text

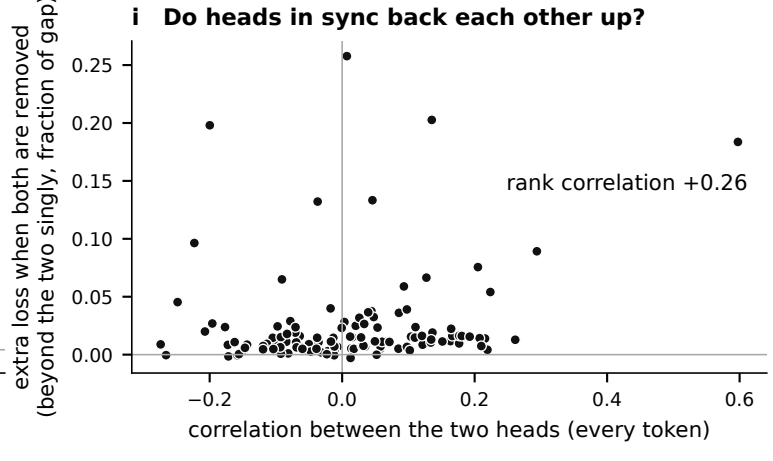

d Without layer norm, averaged over 32 tokens
r(red, black) = 0.48
r(red, grey) = -0.06
head activity (z-score)
token position in the text


e With layer norm
every token, |r| = 0.10
8-token avg, |r| = 0.12
32-token avg, |r| = 0.15


f Without layer norm
every token, |r| = 0.14
8-token avg, |r| = 0.19
32-token avg, |r| = 0.23
correlation between two heads


g Does the same network reappear?
ICC of the network across 4 segments of text
0.92
0.87
0.84
0.73
0.67
0.57
with layer norm
without layer norm
every token
8-token average
32-token average


h Where the sync is
mean correlation (every token)
0.07
0.17
-0.01
0.07
with layer norm
without layer norm
same layer
different layers


i Do heads in sync back each other up?
extra loss when both are removed (beyond the two singly, fraction of gap)
rank correlation +0.26
correlation between the two heads (every token)